

DrainLite V5 evidence supplement

Feature dictionary

Feature	Unit	Calculation	Availability
surface_depth_mm	mm	1000 B(t,x)	Contemporaneous surface-only field
rainfall_mm	mm/5 min	Released rainfall at t,x; not redistributed forcing	Current interval
cumulative_rainfall_mm	mm	Sum of released rainfall through t	Past and current rainfall
dem_m	m	Released DEM at x	Static
dem_slope	m/m	$\sqrt{g_x^2 + g_y^2}$, numpy.gradient with 20 m spacing; wall gradients retained	Static
time_norm	1	(t+1)/72 for zero-based index t	Simulation clock
time_sin	1	$\sin(2 \pi (t+1)/72)$	Simulation clock
time_cos	1	$\cos(2 \pi (t+1)/72)$	Simulation clock
surface_depth_3x3_mean_mm	mm	1000 $K3(B M)/(K3M)$; zero if denominator is zero	Current B
surface_depth_7x7_mean_mm	mm	1000 $K7(B M)/(K7M)$	Current B
drain_inlet_mask	0/1	One in active cells receiving a junction	Static
pipe_mask	0/1	One in active rasterised conduit cells	Static
drain_inlet_count	count	Number of junctions per active cell	Static
pipe_segment_count	count	Number of rasterised conduits visiting cell	Static
network_degree	count	Junction in-degree + out-degree; last junction wins if collocated	Static
pipe_density_3x3	fraction	Uniform 3x3 pipe-mask mean; zero padding	Static
pipe_density_7x7	fraction	Uniform 7x7 pipe-mask mean; zero padding	Static
inlet_density_7x7	fraction	Uniform 7x7 inlet-mask mean; zero padding	Static
pipe_diameter	m	Maximum conduit diameter in cell	Static
pipe_slope	m/m	Arithmetic mean of slopes of visiting conduits, including offsets	Static
pipe_capacity	m ³ /s	Sum of Manning circular full-flow capacities: $A(D/4)^{2/3} \sqrt{S}/n$	Static descriptor, not simulated flow
pipe_cover_depth	m	Mean endpoint cover for two-junction segments; node cover overwrites at junction cells	Static
capacity_density_7x7	m ³ /s	Uniform 7x7 mean of summed capacity; zero padding	Static descriptor

Conduits are sampled between endpoint grid indices using $\max(\text{abs}(\text{delta row}), \text{abs}(\text{delta col}))+1$ linearly spaced points, rounded to nearest integer. Only in-bounds active cells are retained. Empty network cells are zero. Network inputs use `numpy.nan_to_num` with NaN mapped to zero; finite-array checks guard the published input. The surface mean uses mask normalisation, whereas static densities deliberately divide by the full kernel area. These descriptors are spatial summaries, not an alternative pipe solver. The current 2,276 coupled junctions occupy 2,276 distinct cells. The fitted prior is an additional predictor in mm, constructed from other events' C-B at the same cell and time.

Rainfall reconciliation

event	rectangle_km2	active_km2	inactive_proxy_km2	nonfinite_dem_cells
event1	89.60000000000001	42.2108	47.3892	0
event20	89.60000000000001	42.2108	47.3892	0
event65	89.60000000000001	42.2108	47.3892	0
event66	89.60000000000001	42.2108	47.3892	0
event67	89.60000000000001	42.2108	47.3892	0
event68	89.60000000000001	42.2108	47.3892	0
event69	89.60000000000001	42.2108	47.3892	0
event70	89.60000000000001	42.2108	47.3892	0

event	active_mean_total_mm	rectangle_mean_total_mm	raw_m3	active_direct_m3
event1	23.516315495816148	23.204593295834325	2079131.5593067557	992642.4901307963
event20	18.675267531081833	18.669881801719644	1672821.4094340801	788297.9827009891
event65	24.615166811437362	24.171784052939206	2165791.8511433527	1039025.8832442202
event66	25.947239044736296	25.730634950367467	2305464.891552925	1095253.7178695549
event67	30.202772700273187	29.802417700099927	2670296.6259289533	1274883.1978966915
event68	28.64342060733802	28.483451686874364	2552117.271143943	1209061.6985722238
event69	27.525211785950273	27.029411604588468	2421835.279771127	1161861.2096543896
event70	29.01476381097937	28.77255241394908	2578020.6962898374	1224736.392272488

event	inactive_reassigned_m3	injected_reconstructed_m3	B_loss_m3	C_loss_m3
event1	1086489.0691759593	2079131.5473588225	216591.17060327576	216366.78117551716
event20	884523.4267330911	1672821.39928162	215209.63248080222	215018.28906270908
event65	1126765.9678991304	2165791.864224041	243120.1110522032	243058.18999845837
event66	1210211.1736833702	2305464.8872295674	244117.59013165397	244058.1379252334
event67	1395413.428032262	2670296.623203429	247835.51332904352	247748.18573747046
event68	1343055.5725717195	2552117.273145773	247232.47799703796	247142.75892084587
event69	1259974.0701167372	2421835.291888452	246255.4364310414	246149.6843282382
event70	1353284.3040173494	2578020.705474043	247379.13248048295	247285.1227998474

Saved-time surface and exchange ledger

The archive contains cumulative values at 30-minute intervals, not every solver step. The table below gives six-hour totals. Positive forward exchange enters SWMM; positive return exchange leaves SWMM. Surface closure and the difference between independently accumulated exchange integrals must be read separately. No pipe-storage or terminal-discharge time series is invented. The complete saved-time ledger is `water_ledger_saved_times.csv`.

event	scenario	rain_m3	loss_m3	surface_m3	surface_closure_m3
event1	B	2079131.5473588225	216591.17060327576	1862668.5947815287	-128.21802598191425
event1	C	2079131.5473588225	216366.78117551716	1245568.598860283	-128.29953061812557
event20	B	1672821.39928162	215209.63248080222	1457687.5705317194	-75.80373090156354
event20	C	1672821.39928162	215018.28906270908	1103083.2308029418	-71.46177813620307

event	scenario	rain_m3	loss_m3	surface_m3	surface_closure_m3
event65	B	2165791.864224041	243120.1110522032	1922675.4512015388	-3.6980297011323273
event65	C	2165791.864224041	243058.18999845837	1362190.1329108102	0.6535983544308692
event66	B	2305464.8872295674	244117.59013165397	2061350.3922647869	-3.0951668734196573
event66	C	2305464.8872295674	244058.1379252334	1442524.224113583	0.5876900493167341
event67	B	2670296.623203429	247835.51332904352	2422462.282209962	-1.1723355767317116
event67	C	2670296.623203429	247748.18573747046	1499255.3198106063	0.07758186687715352
event68	B	2552117.273145773	247232.47799703796	2304886.810986713	-2.015837978105992
event68	C	2552117.273145773	247142.75892084587	1448200.6791682236	1.1051976804155856
event69	B	2421835.291888452	246255.4364310414	2175582.9065051666	-3.051047755870968
event69	C	2421835.291888452	246149.6843282382	1380076.9851031739	1.6861088073346764
event70	B	2578020.705474043	247379.13248048295	2330642.7524804105	-1.1794868507422507
event70	C	2578020.705474043	247285.1227998474	1460853.7290347114	22.20162365026772

event	scenario	surface_net_exchange_m3	swmm_forward_exchange_m3	swmm_return_exchange_m3	exchange_integral_mismatch_m3
event1	B	0.0	0.0	0.0	0.0
event1	C	617324.4668536405	618893.8989374796	1648.6357824857114	79.20369864662644
event20	B	0.0	0.0	0.0	0.0
event20	C	354791.3411941054	354692.141892141	-1.1641532182693481e-10	99.19930196425412
event65	B	0.0	0.0	0.0	0.0
event65	C	560542.887716418	560418.414221876	0.17786591104231775	124.65136045310646
event66	B	0.0	0.0	0.0	0.0
event66	C	618881.9375007016	618753.4444100781	4.851406619185582	133.34449724270962
event67	B	0.0	0.0	0.0	0.0
event67	C	923293.040073485	925897.3929708564	2696.0548885702156	91.70199119881727
event68	B	0.0	0.0	0.0	0.0
event68	C	856772.7298590228	858945.7825157277	2255.9257858428173	82.87312913790811
event69	B	0.0	0.0	0.0	0.0
event69	C	795606.936348233	797734.7710149152	2199.8739656984108	72.03929901623633
event70	B	0.0	0.0	0.0	0.0
event70	C	869859.6520158339	872112.1639667468	2334.677577877301	82.16562696441542

Across-event and across-seed network gains

event	seed_1907_gain_mm	seed_2718_gain_mm	seed_3141_gain_mm	seed_5772_gain_mm	seed_8119_gain_mm	seed_mean_gain_mm
event1	0.061662912368774414	0.07502555847167969	0.06670951843261719	0.06858062744140625	0.07621407508850098	0.0696385383605957
event20	0.10621047019958496	0.06816840171813965	0.10330891609191895	0.10230183601379395	0.08267855644226074	0.09253363609313965
event65	0.08114886283874512	0.0934000015258789	0.08639407157897949	0.09473586082458496	0.08022761344909668	0.08718128204345703
event66	0.07399272918701172	0.06335806846618652	0.08731555938720703	0.07303524017333984	0.07604694366455078	0.07474970817565918
event67	0.01427769660949707	0.02427387237548828	0.02184581756591797	0.003887176513671875	0.0022933483123779297	0.013315582275390625
event68	0.00024962425231933594	0.00042366981506347656	-0.020188331604003906	0.01455068588256836	-0.0004830360412597656	-0.0010894775390625
event69	0.024358153343200684	0.05244350433496094	0.039286017417907715	0.021840572357177734	0.024727821350097656	0.032531213760375974
event70	0.03266716003417969	0.0005127191543579102	-0.005211830139160156	0.0187680721282959	0.019881248474121094	0.013323473930358886

Signed residual errors

Positive residual is not an independently identified surcharge flow. All counts are cell-times, not unique land area.

event	subset	n_cell_times	active_cell_time_pct	mae_mm	bias_mm
event1	net_negative_lt_minus5mm	654260	8.61101371634221	17.63309097290039	-4.115084171295166
event1	positive_gt5mm	241789	3.1822951050968524	7.202901363372803	-5.639542102813721
event20	net_negative_lt_minus5mm	506889	6.6713968936859755	17.295278549194336	-6.001902103424072
event20	positive_gt5mm	119503	1.572833387558529	9.83563899938965	-5.983149528503418
event65	net_negative_lt_minus5mm	782047	10.292876599248428	14.73000717163086	3.6041407585144043
event65	positive_gt5mm	262475	3.4545529685399106	7.566878318786621	-4.707199573516846
event66	net_negative_lt_minus5mm	817217	10.755764980631602	14.89147663116455	3.636434316635132
event66	positive_gt5mm	275905	3.631311312639314	7.330359935760498	-4.4908270835876465
event67	net_negative_lt_minus5mm	901804	11.869052996442194	17.144874572753906	6.311272144317627
event67	positive_gt5mm	327984	4.316746740960449	6.076767921447754	-4.532792091369629
event68	net_negative_lt_minus5mm	887449	11.680120306230211	12.832428932189941	1.0215444564819336
event68	positive_gt5mm	342616	4.509325154278578	6.36810302734375	-5.397876739501953
event69	net_negative_lt_minus5mm	847254	11.1510956121814	13.484895706176758	1.2359528541564941
event69	positive_gt5mm	290348	3.8214022109138996	4.670828819274902	-2.586766242980957
event70	net_negative_lt_minus5mm	888206	11.690083527859642	12.960941314697266	1.734512448310852
event70	positive_gt5mm	326320	4.2948460794130625	5.650572299957275	-4.421494007110596

event	subset	max_abs_error_mm	wrong_sign_pct
event1	net_negative_lt_minus5mm	1004.3993530273438	3.8108702962125154
event1	positive_gt5mm	234.61962890625	6.436190231979122
event20	net_negative_lt_minus5mm	521.2002563476562	8.904119047759963
event20	positive_gt5mm	266.02587890625	10.97461988401965
event65	net_negative_lt_minus5mm	723.9044189453125	1.7134520048027804
event65	positive_gt5mm	304.66009521484375	8.222116392037337
event66	net_negative_lt_minus5mm	855.2701416015625	1.562008621944967
event66	positive_gt5mm	277.0807800292969	7.8313187510193725
event67	net_negative_lt_minus5mm	1346.8597412109375	0.8119280908046539
event67	positive_gt5mm	265.5171203613281	4.561198107224743
event68	net_negative_lt_minus5mm	1618.5247802734375	0.37647233812872627
event68	positive_gt5mm	189.91741943359375	3.688093959418124
event69	net_negative_lt_minus5mm	1082.66650390625	0.33956759130083775
event69	positive_gt5mm	211.17453002929688	2.514224310138179
event70	net_negative_lt_minus5mm	1072.0374755859375	0.3265008342659248
event70	positive_gt5mm	201.27719116210938	3.190120127482226

Volume trajectories and timing

Final-hour slope is a descriptive slope, not a calibrated recession constant. Rectangular five-minute integration is used for absolute volume-error integrals.

event	model	volume_peak_h	mike_volume_peak_h	peak_time_difference_h	volume_rmse_vs_mike_m3
event1	B	4.416666666666667	2.8333333333333335	1.5833333333333333	340416.1176191825
event1	C	3.0	2.8333333333333335	0.16666666666666666	156872.57960840175
event1	DrainLite	3.5	2.8333333333333335	0.6666666666666666	188298.29848544716
event20	B	5.166666666666667	4.083333333333333	1.0833333333333333	644072.3545487972
event20	C	4.0	4.083333333333333	-0.08333333333333333	495222.1406713802
event20	DrainLite	4.0	4.083333333333333	-0.08333333333333333	461039.78734585305
event65	B	6.0	1.0833333333333333	4.916666666666667	596600.361160304
event65	C	5.25	1.0833333333333333	4.166666666666667	319328.62182566064
event65	DrainLite	5.166666666666667	1.0833333333333333	4.083333333333333	309529.7327028922
event66	B	6.0	1.0833333333333333	4.916666666666667	616656.7760013543
event66	C	5.25	1.0833333333333333	4.166666666666667	324218.3018172328
event66	DrainLite	5.166666666666667	1.0833333333333333	4.083333333333333	305581.44780928444
event67	B	6.0	2.75	3.25	373913.17752543703
event67	C	2.75	2.75	0.0	375816.6700015634
event67	DrainLite	2.75	2.75	0.0	356752.56825873867
event68	B	6.0	2.75	3.25	443360.3167875795
event68	C	2.75	2.75	0.0	245934.28138848665
event68	DrainLite	2.75	2.75	0.0	256247.8840741612
event69	B	6.0	2.75	3.25	433628.1254075166
event69	C	2.75	2.75	0.0	207927.21773436083
event69	DrainLite	2.75	2.75	0.0	212398.05641569436
event70	B	6.0	2.75	3.25	425098.43581838947
event70	C	2.75	2.75	0.0	276086.5341237564
event70	DrainLite	2.75	2.75	0.0	279181.73894799757

event	model	last_hour_volume_slope_m3h	volume_bias_vs_C_mean_m3	volume_error_vs_C_maxabs_m3	volume_error_vs_C_integral_abs_m3h
event1	B	-18086.462229268844	251246.2771034014	617099.9959212418	1507477.6626204085
event1	C	-96399.87440194123	0.0	0.0	0.0
event1	DrainLite	-62534.538895503	-29528.224668664934	88547.9343951107	186311.51479413756
event20	B	-9801.25242055588	112813.6837465064	354604.33972877776	676882.1024790384
event20	C	-96880.9182374914	0.0	0.0	0.0
event20	DrainLite	-74484.98248049365	-28847.677142284185	85121.45123184449	174420.48065232814
event65	B	50615.34710817417	270267.30581327155	560485.3182907286	1621603.8348796293
event65	C	-56366.04642409821	0.0	0.0	0.0
event65	DrainLite	-67631.25861783544	5137.703835714013	38505.99169970909	98518.77982178055
event66	B	56308.536889424555	299112.450954739	618826.1681512038	1794674.705728434
event66	C	-61030.08475269133	0.0	0.0	0.0
event66	DrainLite	-91094.73457050753	6664.114835048525	66564.74793108366	159232.95243476963
event67	B	28257.362145613293	461265.23658821837	923206.9623993558	2767591.41952931
event67	C	-53094.69525489451	0.0	0.0	0.0
event67	DrainLite	-41439.890551325756	20994.526063506302	56802.64370036102	144920.31546042033
event68	B	25276.908469112877	426383.56690574676	856686.1318184894	2558301.4014344807

event	model	last_hour_volume_slope_m3h	volume_bias_vs_C_mean_m3	volume_error_vs_C_maxabs_m3	volume_error_vs_C_integral_abs_m3h
event68	C	-49308.39275459697	0.0	0.0	0.0
event68	DrainLite	-47296.639797252385	-8061.28013343713	26635.608098803554	58839.501167028844
event69	B	22091.246880503775	397608.8413244845	795505.9214019927	2385653.047946907
event69	C	-47097.415580801775	0.0	0.0	0.0
event69	DrainLite	-48608.614513059125	311.6681296685564	26103.338357732166	55278.99417764332
event70	B	25927.849185089235	435564.0102752852	869789.0234456991	2613384.0616517114
event70	C	-47078.758308088465	0.0	0.0	0.0
event70	DrainLite	-48951.05644033392	-1287.5427786911273	16810.790495980764	36147.467888651896

Display bounds and tails

Threshold columns apply to the named field; only rows named error describe prediction-error tails.

event	field	time_h	display_abs_bound_m	clipped_pct	min_m
event1	B	3.0	0.5	0.528774626398931	6.428585766116157e-05
event1	C_minus_B	3.0	0.2	0.35820216627024365	-1.5987751483917236
event1	pred_minus_B	3.0	0.2	0.6794469661792716	-1.281444787979126
event1	error	3.0	0.2	0.084338605285851	-0.35164451599121094
event20	B	4.0	0.5	0.2151108247178447	0.00021290325094014406
event20	C_minus_B	4.0	0.2	0.14309134155239892	-1.1150847673416138
event20	pred_minus_B	4.0	0.2	0.17057246012868746	-1.181192398071289
event20	error	4.0	0.2	0.015161996455883328	-0.4152621626853943
event65	B	5.25	0.5	0.7685236953575862	2.075177326332777e-05
event65	C_minus_B	5.25	0.2	1.6564481128052537	-1.750544786453247
event65	pred_minus_B	5.25	0.2	1.7957489552436818	-1.1740772724151611
event65	error	5.25	0.2	0.00947624778492708	-0.23511876165866852
event66	B	5.25	0.5	0.8159049342822217	2.3660026272409596e-05
event66	C_minus_B	5.25	0.2	1.995697783505643	-1.7951658964157104
event66	pred_minus_B	5.25	0.2	2.1662702436343304	-1.2970112562179565
event66	error	5.25	0.2	0.011371497341912496	-0.2513996660709381
event67	B	2.75	0.5	1.1068257412794829	9.28855879465118e-05
event67	C_minus_B	2.75	0.2	1.5124091464743619	-1.878380298614502
event67	pred_minus_B	2.75	0.2	1.3721606792574412	-1.2981373071670532
event67	error	2.75	0.2	0.030323992911766656	-0.3270007371902466
event68	B	2.75	0.5	0.9106674121314924	8.870556484907866e-05
event68	C_minus_B	2.75	0.2	1.349417684573616	-1.793360948562622
event68	pred_minus_B	2.75	0.2	1.3797416774853828	-1.0836429595947266
event68	error	2.75	0.2	0.020847745126839578	-0.3186982572078705
event69	B	2.75	0.5	0.8462289271939883	8.207270002458245e-05
event69	C_minus_B	2.75	0.2	1.2925601978640537	-1.7843581438064575
event69	pred_minus_B	2.75	0.2	1.1504164810901476	-1.342987298965454
event69	error	2.75	0.2	0.011371497341912496	-0.3389068841934204

event	field	time_h	display_abs_bound_m	clipped_pct	min_m
event70	B	2.75	0.5	0.9438342793787371	8.952220377977937e-05
event70	C_minus_B	2.75	0.2	1.3901655500488026	-1.8154338598251343
event70	pred_minus_B	2.75	0.2	1.4091180456186567	-1.3159208297729492
event70	error	2.75	0.2	0.021795369905332287	-0.31251060962677

event	field	max_m	abs_gt_50mm_pct	abs_gt_100mm_pct	abs_gt_200mm_pct
event1	B	3.19879150390625	24.087674244506147	14.535616477299648	5.031887573796279
event1	C_minus_B	0.15891188383102417	3.7601751210590653	1.5323092668227087	0.35820216627024365
event1	pred_minus_B	0.05044245719909668	5.0934831843983055	2.2259706046793712	0.6794469661792716
event1	error	0.4464743435382843	1.5076710225818986	0.3534640423777801	0.084338605285851
event20	B	2.6056466102600098	21.452329735517925	9.956693547622884	2.5453201550314137
event20	C_minus_B	0.07438862323760986	1.7862727074587545	0.4842362618097738	0.14309134155239892
event20	pred_minus_B	0.030451133847236633	3.1072616486775892	0.8481241767509737	0.17057246012868746
event20	error	0.34396207332611084	0.892662541340131	0.16393908667923848	0.015161996455883328
event65	B	3.4067070484161377	23.348526917281834	15.89735328399367	6.473224861883689
event65	C_minus_B	0.080777607858181	6.79920778568518	4.008452813024155	1.6564481128052537
event65	pred_minus_B	0.03848540782928467	7.08633809356847	4.47563182882106	1.7957489552436818
event65	error	0.6418855786323547	0.8888720422261601	0.19521070436949786	0.00947624778492708
event66	B	3.4788103103637695	24.20897021615321	17.397443308347626	7.630274716423285
event66	C_minus_B	0.09008249640464783	7.328930036862603	4.6111421721455175	1.995697783505643
event66	pred_minus_B	0.04011571407318115	7.872866659717419	5.335127502913947	2.1662702436343304
event66	error	0.6717064380645752	0.8291716811811196	0.19426307959100514	0.011371497341912496
event67	B	3.743831157684326	26.3032209766221	19.143915775109686	8.594009116150369
event67	C_minus_B	0.15973486006259918	7.059804599770675	4.021719559923053	1.5124091464743619
event67	pred_minus_B	0.05499058961868286	6.299809527419523	3.7431178750461966	1.3721606792574412
event67	error	0.6303362846374512	0.9589962758346205	0.19331545481251244	0.030323992911766656
event68	B	3.562718629837036	25.997138173168956	18.611350649596787	8.07660598709335
event68	C_minus_B	0.16452491283416748	6.741402674197125	3.6568840202033606	1.349417684573616
event68	pred_minus_B	0.054155826568603516	6.267590284950771	3.7033176343495033	1.3797416774853828
event68	error	0.7835773229598999	0.5401461237408436	0.14972471500184786	0.020847745126839578
event69	B	3.5369489192962646	25.011608403536535	17.207918352649084	7.256910553697159
event69	C_minus_B	0.16504907608032227	6.038265088555535	3.3422725937437816	1.2925601978640537
event69	pred_minus_B	0.05481690168380737	5.539814455068371	3.3470107176362442	1.1504164810901476
event69	error	0.44443440437316895	0.5031887573796279	0.07580998227941664	0.011371497341912496
event70	B	3.611003875732422	26.061576658106457	18.63030314516664	8.170420840164129
event70	C_minus_B	0.1691642701625824	6.837112776824888	3.7563846219450947	1.3901655500488026
event70	pred_minus_B	0.05429753661155701	6.2401091663744825	3.6559363954248676	1.4091180456186567
event70	error	0.5006462931632996	0.6472277237105196	0.15541046367280412	0.021795369905332287

Conditional subset support

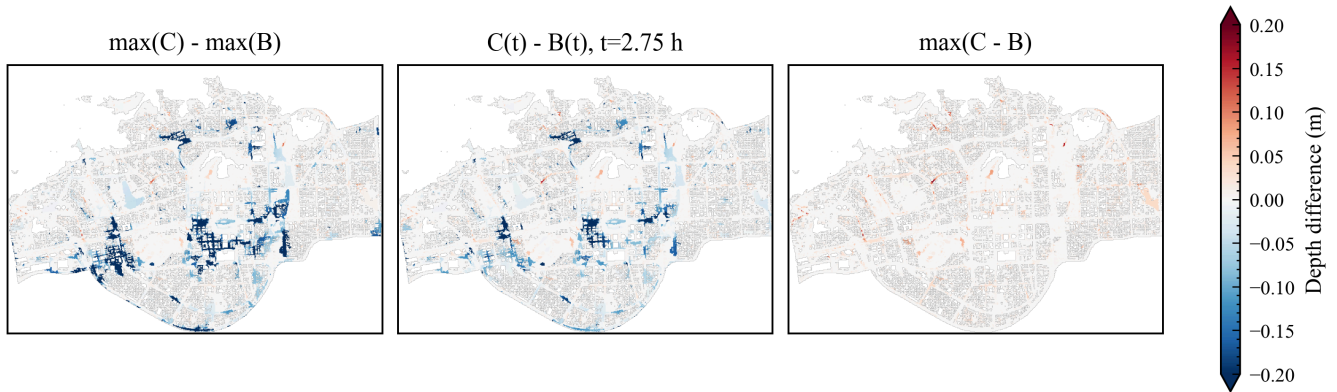
Event	Subset	Cell-times	Active %	MAE mm
event1	all_active	7597944	100.0	2.245790958404541
event1	target_wet_0p03m	1291340	16.995913631371856	10.129962921142578
event1	target_wet_0p15m	304579	4.008702880674035	15.476371765136719
event1	wet_either_0p03m	1409968	18.557230745580647	10.653377532958984
event1	effect_abs_over_1mm	1570071	20.66441921656701	9.755302429199219
event1	net_drainage_over_1mm	963819	12.685260644195324	13.203367233276367
event1	positive_residual_over_1mm	606252	7.979158572371683	4.273573398590088
event1	effect_abs_over_5mm	896049	11.793308821439062	14.818618774414062
event1	net_drainage_over_5mm	654260	8.61101371634221	17.63309097290039
event1	positive_residual_over_5mm	241789	3.1822951050968524	7.202901363372803
event1	effect_abs_over_10mm	661920	8.71183046360963	18.044424057006836
event1	net_drainage_over_10mm	539221	7.096933065050229	19.863191604614258
event1	positive_residual_over_10mm	122699	1.6148973985593997	10.051538467407227
event1	near_inlet_0_20m	163872	2.1567939958494033	4.218630313873291
event1	near_inlet_20_60m	1650312	21.72050754783136	3.5119645595550537
event1	near_inlet_60_100m	1296936	17.06956513498915	2.6845054626464844
event1	near_inlet_100_infm	4486824	59.05313332133009	1.5812097787857056
event1	near_pipe_0_20m	589536	7.759151686298293	4.043787479400635
event1	near_pipe_20_60m	1712232	22.53546485733509	3.1739261150360107
event1	near_pipe_60_100m	1041552	13.708340045675515	2.487098217010498
event1	near_pipe_100_infm	4254624	55.997043410691106	1.5640619993209839
event20	all_active	7597944	100.0	2.1207780838012695
event20	target_wet_0p03m	1099047	14.465057915667712	10.777276992797852
event20	target_wet_0p15m	157302	2.070323234811944	19.107192993164062
event20	wet_either_0p03m	1222127	16.08496982867997	10.792914390563965
event20	effect_abs_over_1mm	1298061	17.084371772153098	10.179698944091797
event20	net_drainage_over_1mm	868164	11.426301641601992	12.469077110290527
event20	positive_residual_over_1mm	429897	5.658070130551107	5.556365013122559
event20	effect_abs_over_5mm	626392	8.244230281244505	15.8721284866333
event20	net_drainage_over_5mm	506889	6.671396893685976	17.295278549194336
event20	positive_residual_over_5mm	119503	1.572833387558529	9.835639953613281
event20	effect_abs_over_10mm	424374	5.585379413167562	18.81374740600586
event20	net_drainage_over_10mm	373891	4.920949667436348	19.534961700439453
event20	positive_residual_over_10mm	50483	0.6644297457312136	13.472233772277832
event20	near_inlet_0_20m	163872	2.1567939958494033	3.256046772003174
event20	near_inlet_20_60m	1650312	21.72050754783136	3.1542341709136963
event20	near_inlet_60_100m	1296936	17.06956513498915	2.5145928859710693
event20	near_inlet_100_infm	4486824	59.05313332133009	1.585362434387207
event20	near_pipe_0_20m	589536	7.759151686298293	3.407017469406128
event20	near_pipe_20_60m	1712232	22.53546485733509	2.931978702545166

Event	Subset	Cell-times	Active %	MAE mm
event20	near_pipe_60_100m	1041552	13.708340045675515	2.3243672847747803
event20	near_pipe_100_infin	4254624	55.997043410691106	1.5662530660629272
event65	all_active	7597944	100.0	2.217952013015747
event65	target_wet_0p03m	1588505	20.907037482771656	7.78990364074707
event65	target_wet_0p15m	313220	4.1224310155484165	13.411247253417969
event65	wet_either_0p03m	1735210	22.837888776226833	8.691130638122559
event65	effect_abs_over_1mm	1880287	24.747313220523868	8.272703170776367
event65	net_drainage_over_1mm	1169781	15.396020291805257	10.730280876159668
event65	positive_residual_over_1mm	710506	9.35129292871861	4.226531028747559
event65	effect_abs_over_5mm	1044522	13.74742956778834	12.930004119873047
event65	net_drainage_over_5mm	782047	10.292876599248428	14.730008125305176
event65	positive_residual_over_5mm	262475	3.45455296853991	7.566878795623779
event65	effect_abs_over_10mm	768200	10.11062992830692	15.872427940368652
event65	net_drainage_over_10mm	648117	8.530162896699423	16.735530853271484
event65	positive_residual_over_10mm	120083	1.580467031607498	11.214061737060547
event65	near_inlet_0_20m	163872	2.1567939958494033	4.857062339782715
event65	near_inlet_20_60m	1650312	21.72050754783136	3.5494747161865234
event65	near_inlet_60_100m	1296936	17.06956513498915	2.602726936340332
event65	near_inlet_100_infin	4486824	59.05313332133009	1.5205916166305542
event65	near_pipe_0_20m	589536	7.759151686298293	4.210177898406982
event65	near_pipe_20_60m	1712232	22.53546485733509	3.161874294281006
event65	near_pipe_60_100m	1041552	13.708340045675515	2.4683852195739746
event65	near_pipe_100_infin	4254624	55.997043410691106	1.5007226467132568
event66	all_active	7597944	100.0	2.292654037475586
event66	target_wet_0p03m	1642467	21.617255931341425	7.86324405670166
event66	target_wet_0p15m	345695	4.549849275014399	12.230406761169434
event66	wet_either_0p03m	1788417	23.538170326077687	8.747039794921875
event66	effect_abs_over_1mm	1938104	25.508269079108768	8.342324256896973
event66	net_drainage_over_1mm	1206081	15.873781117628663	10.917603492736816
event66	positive_residual_over_1mm	732023	9.634487961480106	4.0992937088012695
event66	effect_abs_over_5mm	1093122	14.387076293270917	12.983043670654297
event66	net_drainage_over_5mm	817217	10.755764980631602	14.89147663116455
event66	positive_residual_over_5mm	275905	3.631311312639314	7.3303608894348145
event66	effect_abs_over_10mm	810077	10.661792190097742	15.922419548034668
event66	net_drainage_over_10mm	682107	8.977520760879521	16.879789352416992
event66	positive_residual_over_10mm	127970	1.6842714292182202	10.819449424743652
event66	near_inlet_0_20m	163872	2.1567939958494033	5.258212566375732
event66	near_inlet_20_60m	1650312	21.72050754783136	3.6900722980499268
event66	near_inlet_60_100m	1296936	17.06956513498915	2.7230303287506104
event66	near_inlet_100_infin	4486824	59.05313332133009	1.5459524393081665
event66	near_pipe_0_20m	589536	7.759151686298293	4.502488613128662
event66	near_pipe_20_60m	1712232	22.53546485733509	3.2666893005371094

Event	Subset	Cell-times	Active %	MAE mm
event66	near_pipe_60_100m	1041552	13.708340045675515	2.584455728530884
event66	near_pipe_100_infin	4254624	55.997043410691106	1.5230257511138916
event67	all_active	7597944	100.0	2.559483528137207
event67	target_wet_0p03m	1636952	21.544670505599935	8.47226619720459
event67	target_wet_0p15m	487406	6.414972260916901	12.936527252197266
event67	wet_either_0p03m	1811312	23.83950184418311	10.005597114562988
event67	effect_abs_over_1mm	1993035	26.23124097782242	9.348690032958984
event67	net_drainage_over_1mm	1229454	16.181403811346858	13.001925468444824
event67	positive_residual_over_1mm	763581	10.049837166475562	3.4665558338165283
event67	effect_abs_over_5mm	1229788	16.185799737402643	14.193014144897461
event67	net_drainage_over_5mm	901804	11.869052996442196	17.144872665405273
event67	positive_residual_over_5mm	327984	4.31674674096045	6.076767444610596
event67	effect_abs_over_10mm	946161	12.452855667269988	17.47842025756836
event67	net_drainage_over_10mm	770011	10.134465323777064	19.527130126953125
event67	positive_residual_over_10mm	176150	2.318390343492924	8.522809982299805
event67	near_inlet_0_20m	163872	2.1567939958494033	5.673966407775879
event67	near_inlet_20_60m	1650312	21.72050754783136	4.24412727355957
event67	near_inlet_60_100m	1296936	17.06956513498915	2.9656763076782227
event67	near_inlet_100_infin	4486824	59.05313332133009	1.7086879014968872
event67	near_pipe_0_20m	589536	7.759151686298293	4.975330352783203
event67	near_pipe_20_60m	1712232	22.53546485733509	3.7550177574157715
event67	near_pipe_60_100m	1041552	13.708340045675515	2.7146687507629395
event67	near_pipe_100_infin	4254624	55.997043410691106	1.7056137323379517
event68	all_active	7597944	100.0	2.047645330429077
event68	target_wet_0p03m	1622214	21.35069697802458	7.058807849884033
event68	target_wet_0p15m	467690	6.155481009072981	10.6428861618042
event68	wet_either_0p03m	1791314	23.57629906195676	7.936159133911133
event68	effect_abs_over_1mm	1990293	26.195152267508156	7.400717258453369
event68	net_drainage_over_1mm	1214510	15.984719023988594	9.78325366973877
event68	positive_residual_over_1mm	775783	10.210433243519562	3.670788526535034
event68	effect_abs_over_5mm	1230065	16.18944546050879	11.031888961791992
event68	net_drainage_over_5mm	887449	11.680120306230211	12.832429885864258
event68	positive_residual_over_5mm	342616	4.509325154278579	6.36810302734375
event68	effect_abs_over_10mm	944501	12.431007651543629	13.407852172851562
event68	net_drainage_over_10mm	754442	9.929554626883272	14.545117378234863
event68	positive_residual_over_10mm	190059	2.5014530246603557	8.89346981048584
event68	near_inlet_0_20m	163872	2.1567939958494033	4.969078063964844
event68	near_inlet_20_60m	1650312	21.72050754783136	3.3390626907348633
event68	near_inlet_60_100m	1296936	17.06956513498915	2.3492648601531982
event68	near_inlet_100_infin	4486824	59.05313332133009	1.3787617683410645
event68	near_pipe_0_20m	589536	7.759151686298293	4.178565979003906
event68	near_pipe_20_60m	1712232	22.53546485733509	2.933948040008545

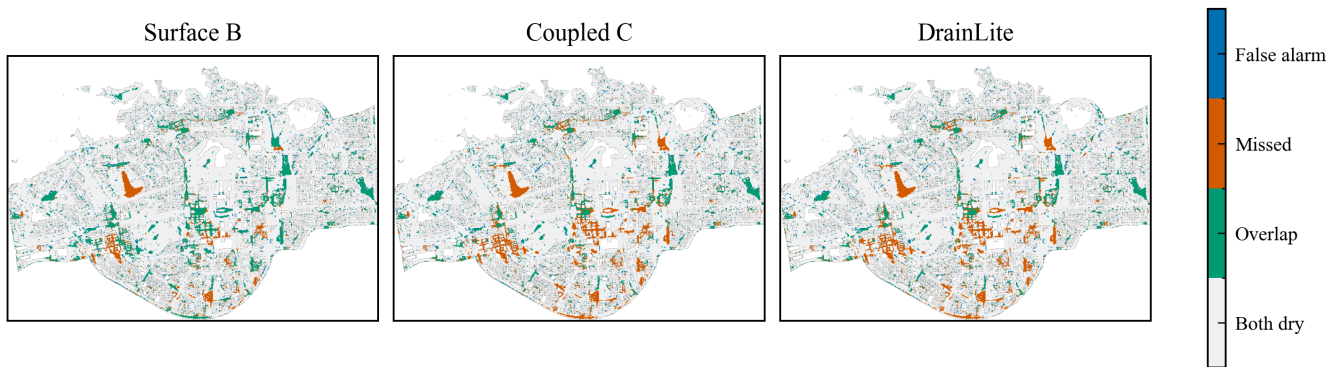
Event	Subset	Cell-times	Active %	MAE mm
event68	near_pipe_60_100m	1041552	13.708340045675515	2.1610500812530518
event68	near_pipe_100_infin	4254624	55.997043410691106	1.367931604385376
event69	all_active	7597944	100.0	1.9361737966537476
event69	target_wet_0p03m	1576342	20.74695470248267	6.794119834899902
event69	target_wet_0p15m	433909	5.710873889041562	10.109517097473145
event69	wet_either_0p03m	1742764	22.937310409237025	7.751852989196777
event69	effect_abs_over_1mm	1891308	24.892365618909537	7.329116344451904
event69	net_drainage_over_1mm	1178778	15.514433904751074	10.114795684814453
event69	positive_residual_over_1mm	712530	9.377931714158462	2.7206149101257324
event69	effect_abs_over_5mm	1137602	14.972497823095301	11.235299110412598
event69	net_drainage_over_5mm	847254	11.1510956121814	13.484896659851074
event69	positive_residual_over_5mm	290348	3.8214022109138996	4.670828819274902
event69	effect_abs_over_10mm	867455	11.41697016982489	13.869034767150879
event69	net_drainage_over_10mm	715416	9.415915674029712	15.437742233276367
event69	positive_residual_over_10mm	152039	2.001054495795178	6.487521171569824
event69	near_inlet_0_20m	163872	2.1567939958494033	4.626025676727295
event69	near_inlet_20_60m	1650312	21.72050754783136	3.355694055557251
event69	near_inlet_60_100m	1296936	17.06956513498915	2.338989019393921
event69	near_inlet_100_infin	4486824	59.05313332133009	1.1993792057037354
event69	near_pipe_0_20m	589536	7.759151686298293	4.025074005126953
event69	near_pipe_20_60m	1712232	22.53546485733509	2.9490644931793213
event69	near_pipe_60_100m	1041552	13.708340045675515	2.1530003547668457
event69	near_pipe_100_infin	4254624	55.997043410691106	1.1860201358795166
event70	all_active	7597944	100.0	2.0126914978027344
event70	target_wet_0p03m	1622383	21.352921263962987	6.829821586608887
event70	target_wet_0p15m	470660	6.194570531185805	10.609872817993164
event70	wet_either_0p03m	1793676	23.607386419273425	7.804935455322266
event70	effect_abs_over_1mm	1971064	25.94207064437432	7.349786758422852
event70	net_drainage_over_1mm	1215624	15.999380885144719	9.890152931213379
event70	positive_residual_over_1mm	755440	9.942689759229602	3.2619316577911377
event70	effect_abs_over_5mm	1214526	15.984929607272704	10.996785163879395
event70	net_drainage_over_5mm	888206	11.690083527859642	12.960941314697266
event70	positive_residual_over_5mm	326320	4.2948460794130625	5.650571823120117
event70	effect_abs_over_10mm	933618	12.287771533983404	13.40543270111084
event70	net_drainage_over_10mm	756569	9.957549042214579	14.68237590789795
event70	positive_residual_over_10mm	177049	2.3302224917688257	7.948777675628662
event70	near_inlet_0_20m	163872	2.1567939958494033	4.848456382751465
event70	near_inlet_20_60m	1650312	21.72050754783136	3.3318636417388916
event70	near_inlet_60_100m	1296936	17.06956513498915	2.357954740524292
event70	near_inlet_100_infin	4486824	59.05313332133009	1.324112892150879
event70	near_pipe_0_20m	589536	7.759151686298293	4.115680694580078
event70	near_pipe_20_60m	1712232	22.53546485733509	2.939002275466919

Event	Subset	Cell-times	Active %	MAE mm
event70	near_pipe_60_100m	1041552	13.708340045675515	2.1506824493408203
event70	near_pipe_100_infm	4254624	55.997043410691106	1.3147281408309937



Supplementary Figure S1. Only the middle panel is an instantaneous C-B field. All use fixed $\pm 0.2 \text{ m}$ display limits; extreme values remain in statistics.

event68: common MIKE-volume-peak time 2.75 h; threshold 0.15 m



Supplementary Figure S2. Event68 at the MIKE surface-volume peak. Green is shared inundation, orange is missed relative to MIKE, blue is false alarm and pale grey is jointly dry. These are disagreements with a model, not observed flooding errors.

Implementation provenance

The 1D/2D coupled hydrodynamic model uses Itzi 25.4, SWMM 5.2.4 and PySWMM 2.1.0. These identify the numerical implementation and calling interface, not separate experimental models. No engine or coupling equation was changed in the editorial revision.

Complete readable-event inventory

Event	Paired A/B/C	Mean 6-h rain (mm)	MIKE maximum (m)
event1	True	23.516	2.708
event20	True	18.675	1.837
event65	True	24.615	2.204
event66	True	25.947	2.228

Event	Paired A/B/C	Mean 6-h rain (mm)	MIKE maximum (m)
event67	True	30.203	3.573
event68	True	28.643	2.795
event69	True	27.525	2.971
event70	True	29.015	2.976
event71	False	16.875	2.450
event72	False	16.004	2.223
event73	False	16.212	2.221
event74	False	23.020	2.010
event75	False	16.727	1.773
event76	False	22.122	1.953
event77	False	23.334	3.905
event78	False	22.130	3.154
event80	False	22.417	3.337

Numerical screening and methodological controls (V7)

Table 1. Physical quality of the eight paired events.

Event	Rain volume (10 ⁶ m3)	B-A MAE (mm)	C-B MAE (mm)	Mean C-B (mm)	Routing continuity (%)	Non-converging (%)	Combined mass error (% rain)	Status
event1	2.079	0.168	7.200	-5.952	-0.083	0.020	-0.100	Accepted
event20	1.673	0.101	3.322	-2.673	-0.178	0.000	-0.032	Accepted
event65	2.166	0.170	7.696	-6.403	-0.111	0.000	-0.017	Accepted
event66	2.305	0.181	8.431	-7.086	-0.101	0.000	-0.015	Accepted
event67	2.670	0.255	12.631	-10.928	-0.066	0.010	-0.111	Accepted
event68	2.552	0.236	11.914	-10.101	-0.060	0.040	-0.097	Accepted
event69	2.422	0.224	10.912	-9.420	-0.068	0.010	-0.101	Accepted
event70	2.578	0.240	12.023	-10.319	-0.063	0.010	-0.098	Accepted

Table 3. Inference-time perturbations of the fixed fitted hybrid; not retraining gains.

Inference-only perturbation	Runs	Mean penalty (mm)	SD across runs (mm)
zero_static_network_fields	8	0.1962	0.1074
shift_20m	32	0.0438	0.0109
shift_40m	32	0.0617	0.0169
shift_80m	32	0.0949	0.0290
shift_160m	32	0.1389	0.0503
block_shuffle	80	0.1924	0.0836
group_permutation	40	0.1610	0.0473

Table 5. Runtime scope and post-processing, canonical seed 1907.

Quantity	Value	Scope
Feature / assembly / estimator / postprocess (s)	3.45 / 1.56 / 18.36 / 0.22	Canonical seed 1907; mean of eight events
B / correction / B+correction / C (s)	274.03 / 23.64 / 297.67 / 3817.57	Historical physical runs + in-memory correction; excludes I/O
Mean event speed ratio; ratio of means	13.098; 12.825	Event-ratio SD 2.841; not integrated deployment timing
Residual bound none: MAE / depth-clamped cells	2.1791 mm / 1.829%	Canonical seed only; fraction over active cell-times
Residual bound 500.0: MAE / depth-clamped cells	2.2104 mm / 1.821%	Canonical seed only; fraction over active cell-times
Residual bound 1200.0: MAE / depth-clamped cells	2.1796 mm / 1.829%	Canonical seed only; fraction over active cell-times

Numerical quality assessment

Global routing continuity errors range from -0.178% to -0.060%, and reported nonconverging steps from 0.00% to 0.04%. Continuity error is a numerical water-budget residual, not a disconnected-pipe count. A nonzero residual is not automatically a model failure, but small global values do not establish local accuracy. In event68 the reported node N01520 has a 63.01% inflow-normalised local imbalance. Its rounded cumulative inflow is about 114 m³, corresponding to approximately 71.8 m³ of imbalance; this is not 63% of whole-network water. The same node's inflow table reports 170.331% because that table uses outflow as its denominator. These are different normalisations of the same budget discrepancy, not conflicting results. The existing aggregate screening therefore does not close the numerical validation. The reported local errors must be checked against node-level exchanges and shorter-step results before the labels can be described as locally validated.

Event	Global balance (%)	Nonconverging steps (%)	Top local node	Local balance (% inflow)	Assessment
event1	-0.083	0.02	N00184	47.07	Local review required
event20	-0.178	0.0	N00184	69.6	Local review required
event65	-0.111	0.0	N00184	65.45	Local review required
event66	-0.101	0.0	N00184	58.75	Local review required
event67	-0.066	0.01	N01520	65.04	Local review required
event68	-0.06	0.04	N01520	63.01	Local review required
event69	-0.068	0.01	N01520	52.93	Local review required
event70	-0.063	0.01	N01520	56.86	Local review required

Interpretation and required follow-up

The EPA manual (Section 8.5) treats continuity as a numerical diagnostic, calls for investigation of excessive local errors at relevant nodes, and recommends time series at one-minute or shorter resolution for initial instability screening. Its example of a 10% excessive global error is not a universal safe threshold. The repository limits of 2% routing/nonconvergence and 0.5% combined error are project screens, not proof of physical validity. A printed 0.00% is rounded and does not prove zero failed iterations.

A nonconverging-step percentage measures steps that fail the iteration stopping criterion; it is neither a percent of wrong depths nor an estimate of mass loss. The current report prints two decimal places and cannot recover exact failed-step counts or their timing.

Next acceptance: instrument inflow, outflow, storage and signed exchange at the flagged nodes; repeat event68 and the event with the largest local discrepancy with reduced maximum routing steps (0.5, 0.25, 0.125 s) and reviewed head tolerance/iteration settings; compare node hydrographs, local budgets and surface labels. Do not alter the physical network merely to improve a diagnostic. No convergence rerun was executed by this audit, and no label or learned prediction has been replaced.

Sources

- EPA manual, Sections 8.5 and 9.1
- Engine 5.2.4 local ranking formula
- Engine 5.2.4 inflow-table formula